LAB EXPERIMENTS

ITA0502 -COMPUTER VISION

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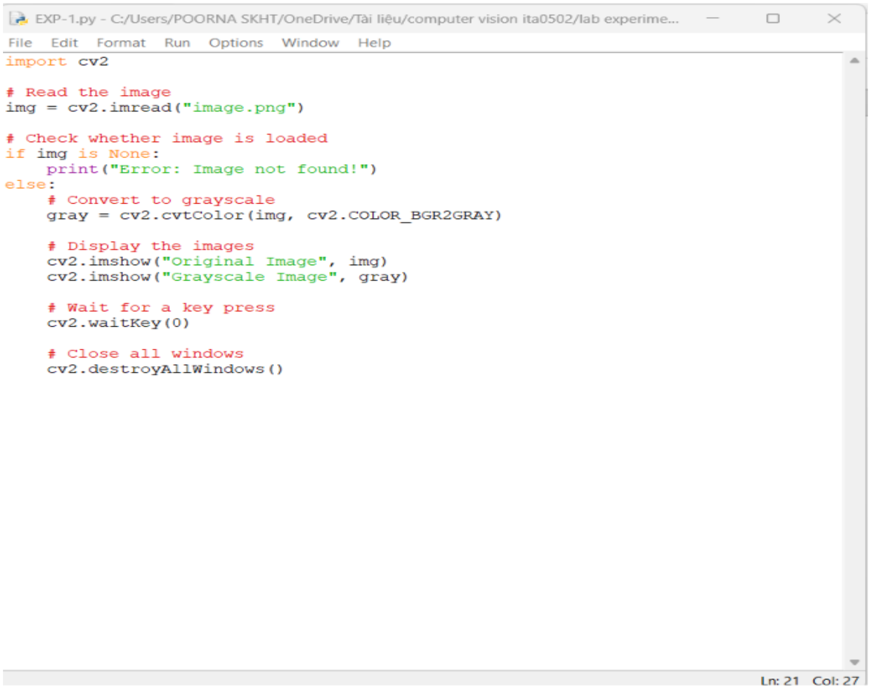
EXP-1

Aim

To read an image using Python and OpenCV, convert the image into grayscale, and display both the original and grayscale images.

Algorithm

1. Import the OpenCV library.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Convert the image to grayscale using cv2.cvtColor().
6. Display the grayscale image.
7. Save the grayscale image.
8. Close all image display windows
9. CODE




```
import cv2

# Read the image
img = cv2.imread("image.png")

# Check whether image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Display the images
    cv2.imshow("Original Image", img)
    cv2.imshow("Grayscale Image", gray)

    # Wait for a key press
    cv2.waitKey(0)

    # Close all windows
    cv2.destroyAllWindows()
```

OUTPUT:

Result

Thus, the input image was successfully read and converted into a grayscale image using Python and OpenCV. The original image and the grayscale image were displayed, and the grayscale image was saved successfully.